

CAR RENTAL SERVICE

SQL FINAL PROJECT BY BIDISHA BHATTACHARYA

CASE STUDY

The purpose of this case-study is research and analysis of a public dataset that is of importance from business point of view. For that I have chosen a car rental business that provides service in the US.

This case study has two datasets. The primary dataset has 5851 records of individual cars and 15 columns that are relevant to car rental business like car owner's ID, car make , model ,kinds of fuel used by the cars, daily fare and the location from where the car is rented. The second dataset is a regional and zonal division of the 51 states of US.

GENERAL OVERVIEW

- 3093 individual car owners or investors
- 51 different car brands are enlisted
- 520 different models enlisted
- 431 cars are listed that has not been rented at all.
- Total revenue is 15 million (15,902,061) U.S.D collected over 1 month.
- Spanned over 46 states

Top Ranked Data Science & Analytics Education Provider since 2007

FACTORS TO OPTIMIZE BUSINESS

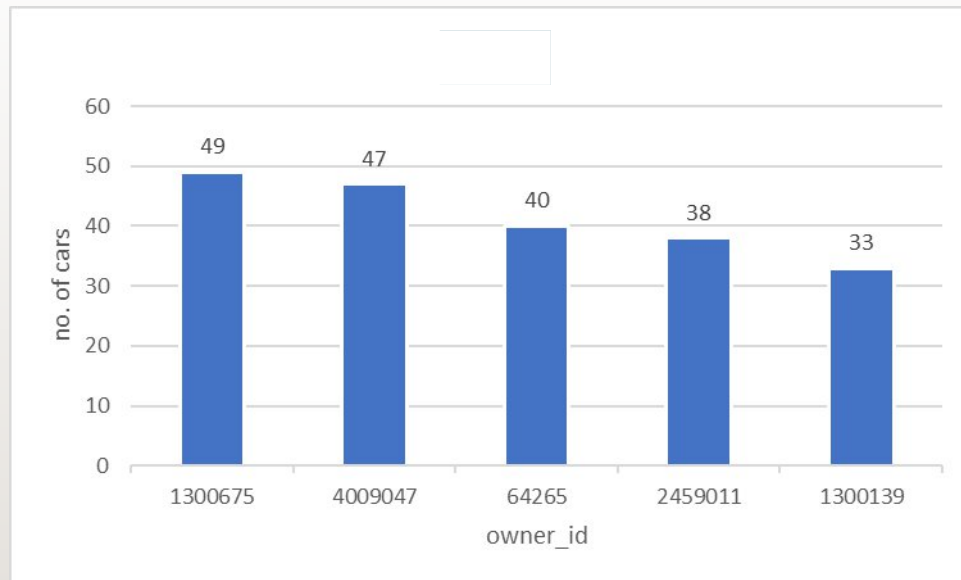
The questions asked here are

- Top investors vs. top earners.
- Most and least favoured car brands that are in circulation.
- Most and least cost effective brand in circulation.
- Most and least effective car based on fuel utilization.
- Car ratings and feedback
- City , State and region wise rentals and revenue.
- State wise distribution of Preferred car make, model and type.
- Growth opportunities in airport cities.

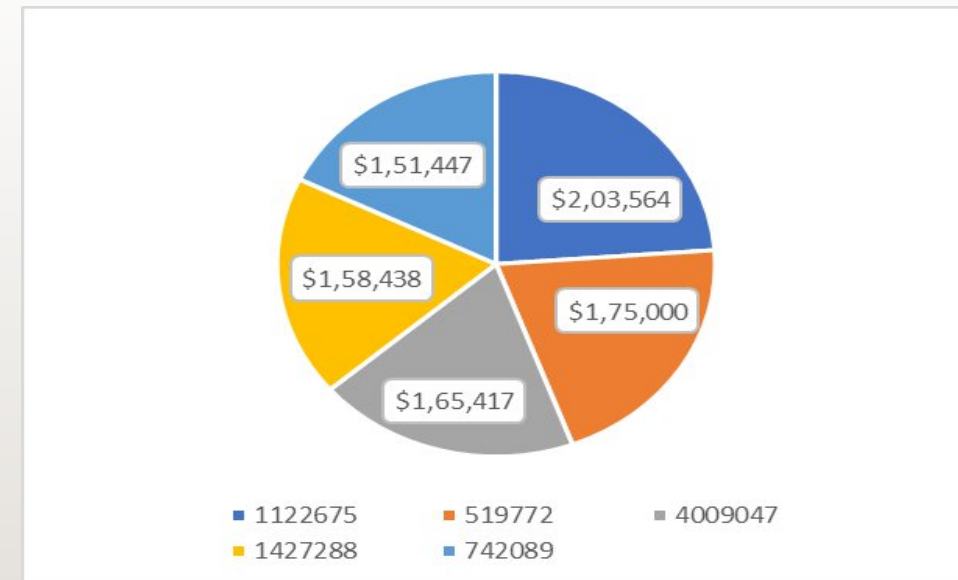
Top Ranked Data Science & Analytics Education Provider since 2007

TOP INVESTORS VS. TOP EARNERS

NO. OF CARS INVESTED

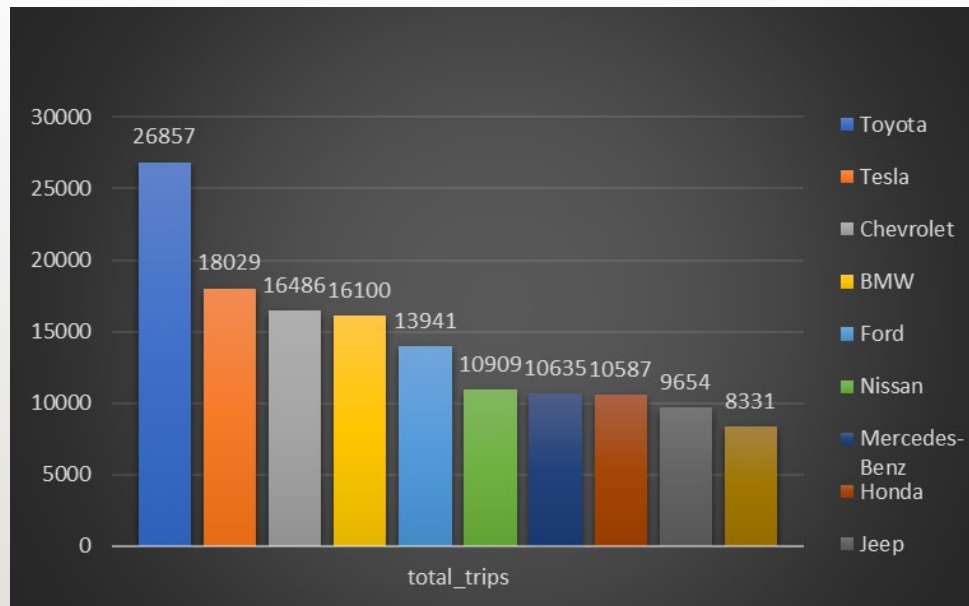


MONEY EARNED BY DIFFERENT OWNERS

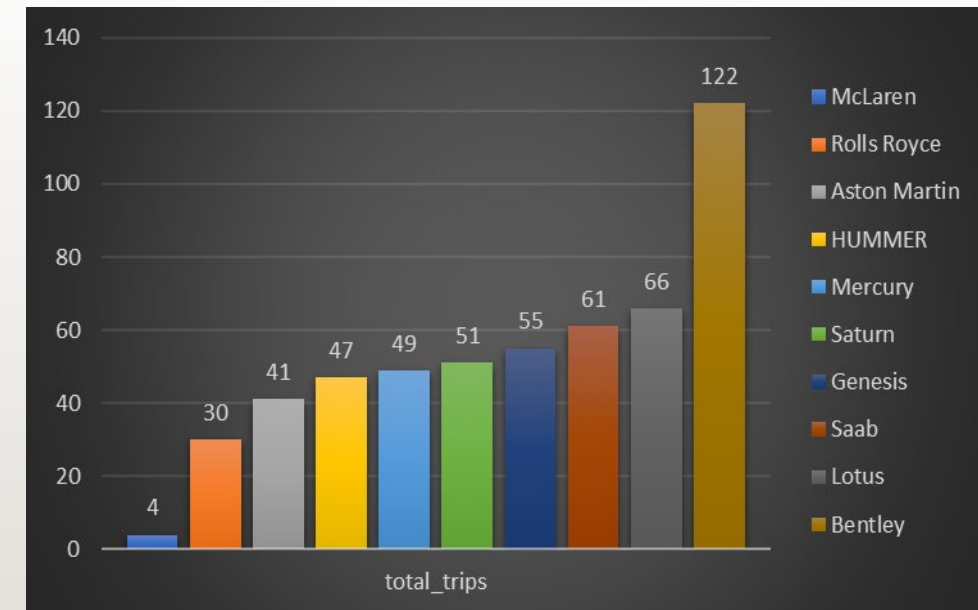


CAR BRANDS PREFERRED BY CLIENTS (IN TERMS OF TOTAL TRIPS TAKEN)

MOST PREFERRED BRANDS

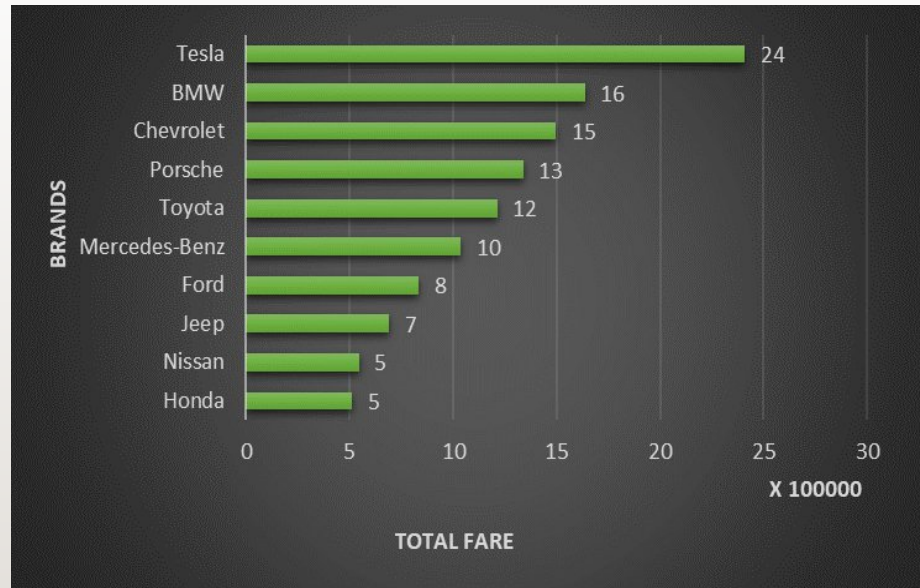


LEAST PREFERRED BRANDS



CAR BRANDS IN TERMS OF COST EFFECTIVENESS

MOST EFFECTIVE



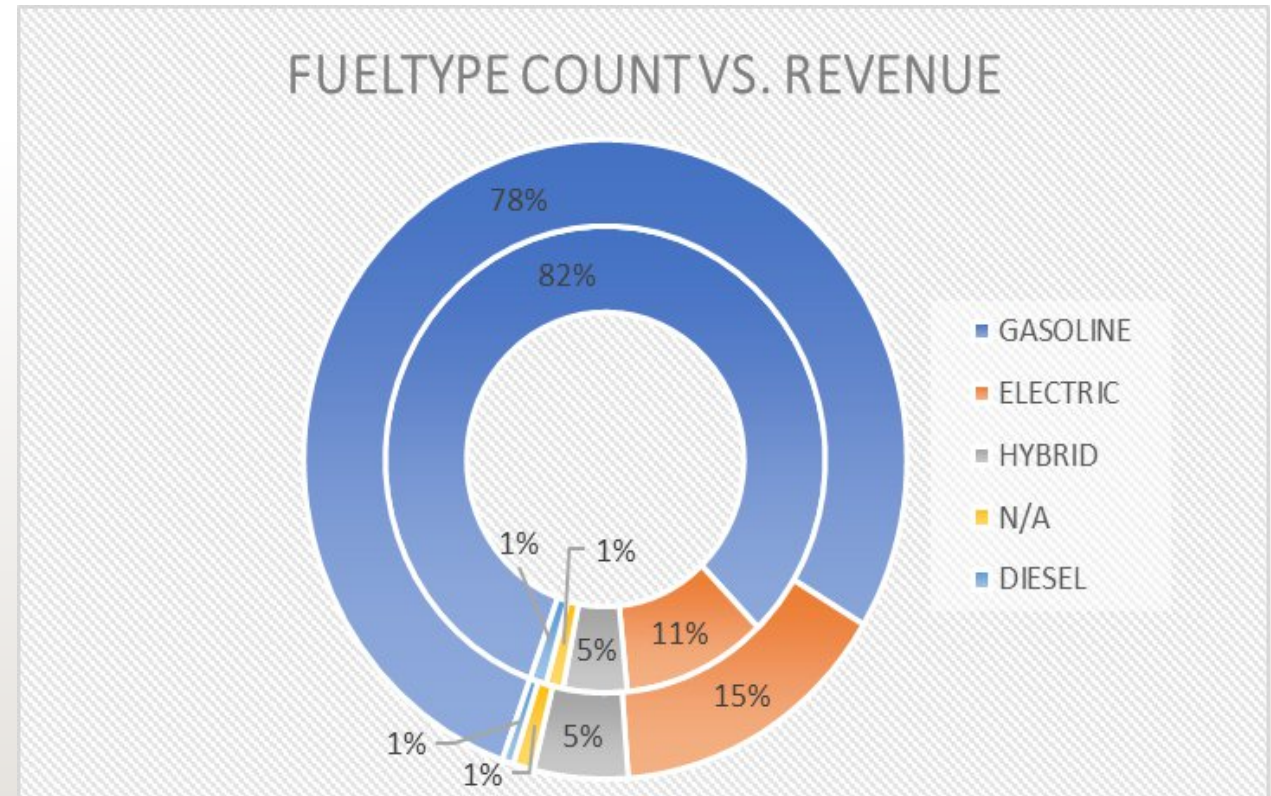
LEAST EFFECTIVE



CARS BASED ON FUEL UTILIZATION

Representing the number of cars using different fuel (inner chart) and the revenue(outer chart) earned from them.

Tesla earns the most revenue yet fuel market is predominantly gasoline.

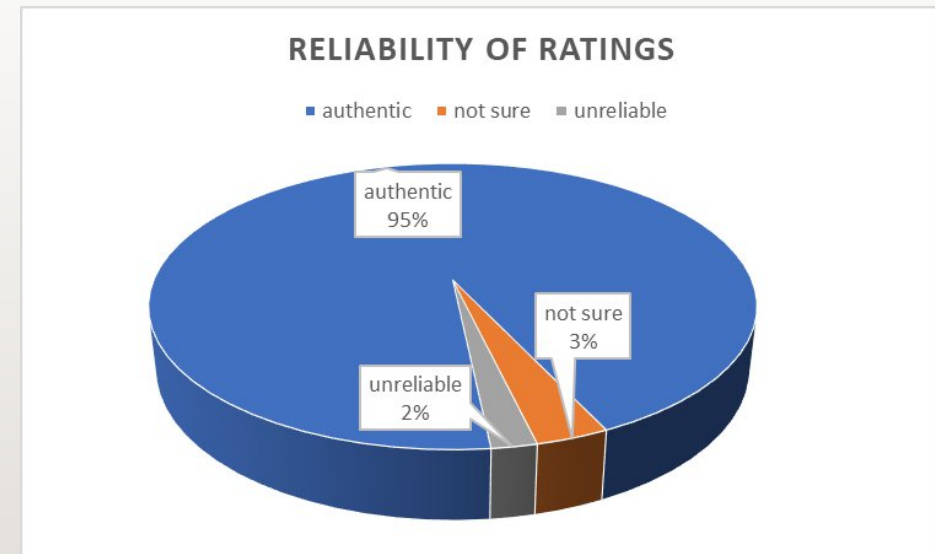


CAR MODELS AND RATINGS (OVERALL FEEDBACK IS SATISFACTORY)

CARS ARE CATEGORIZED INTO BEST, GOOD AND BAD BASED ON GIVEN RATINGS.



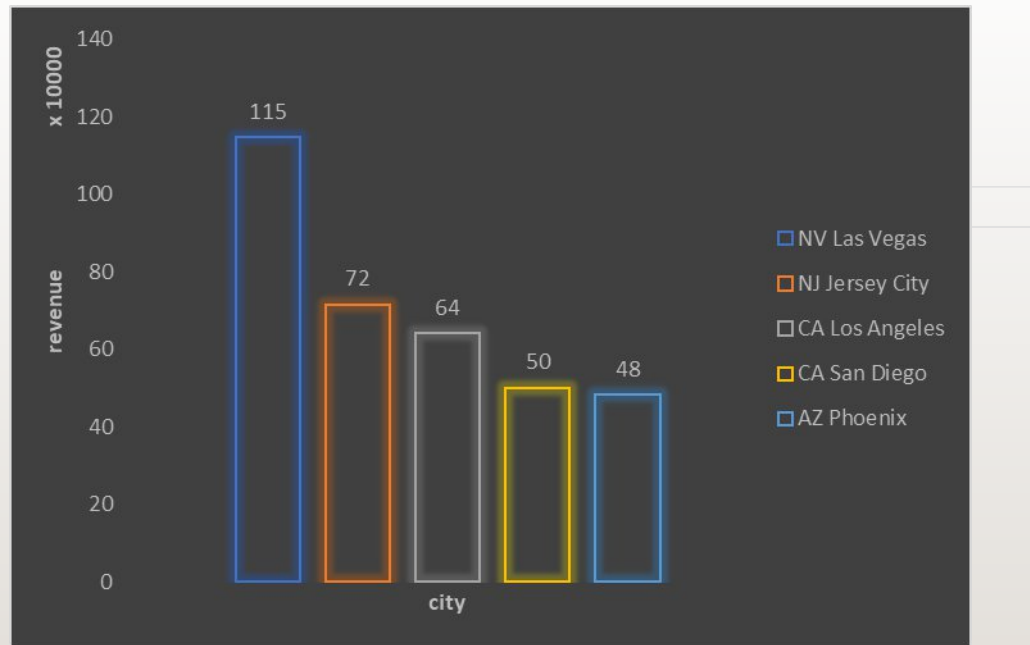
AUTHENTICITY OF THE RATINGS ASSIGNED
BASED ON NO. OF MISSING RATINGS.



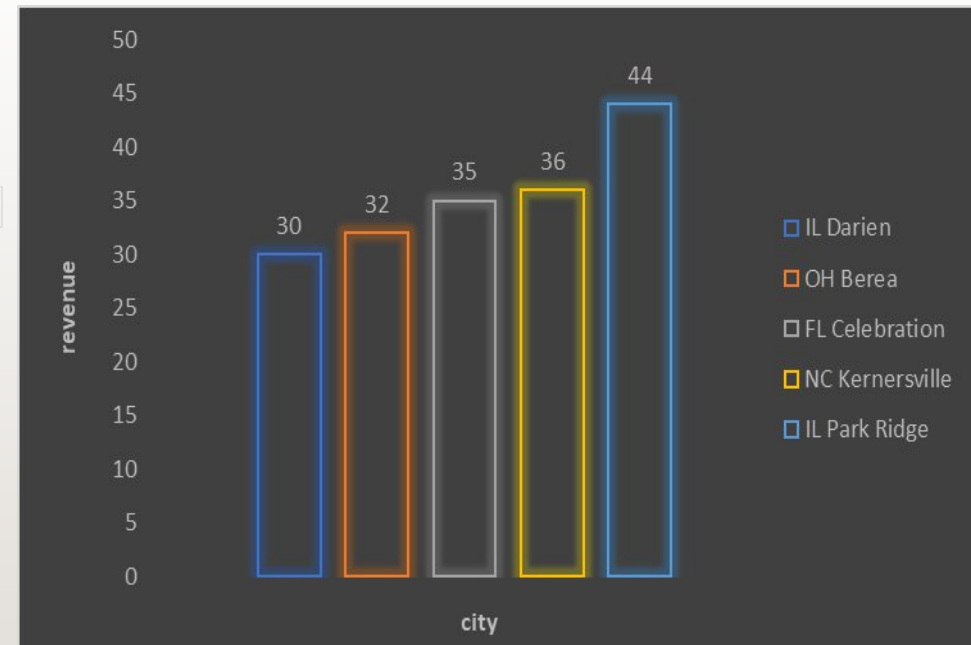
CITY WISE REVENUE

(EXCLUDING 78 CITIES WITH ZERO RENTALS)

TOP 5 CITIES



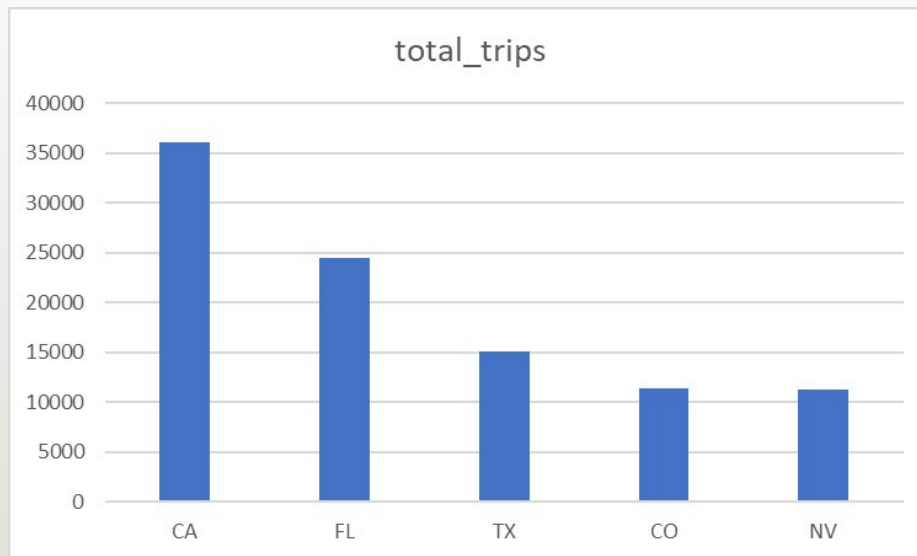
BOTTOM 5 CITIES



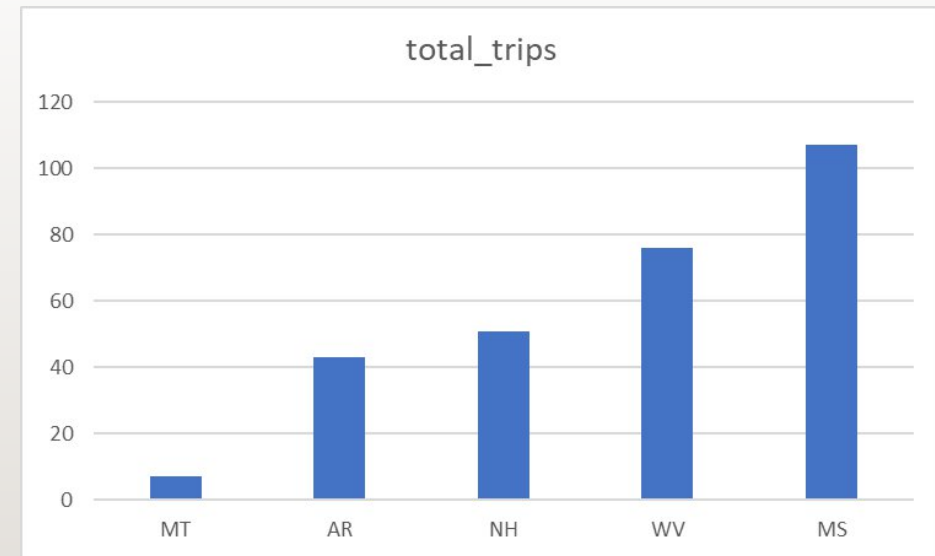
Top Ranked Data Science & Analytics Education Provider since 2007

STATES IN TERMS OF TOTAL RENTALS

TOP 5 STATES



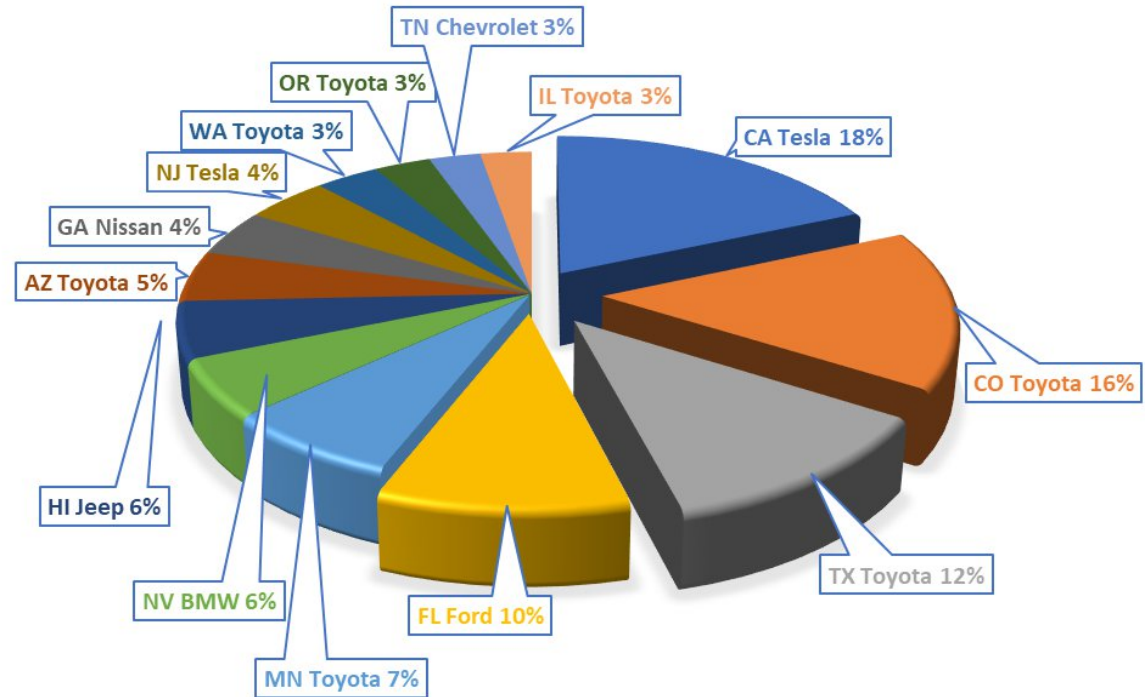
BOTTOM 5 STATES



Top Ranked Data Science & Analytics Education Provider since 2007

TOP BRANDS IN EACH STATE

- Tesla is a favourite in California.
- Toyota is a favourite in most of the states viz. Colorado , Texas , Minnesota.

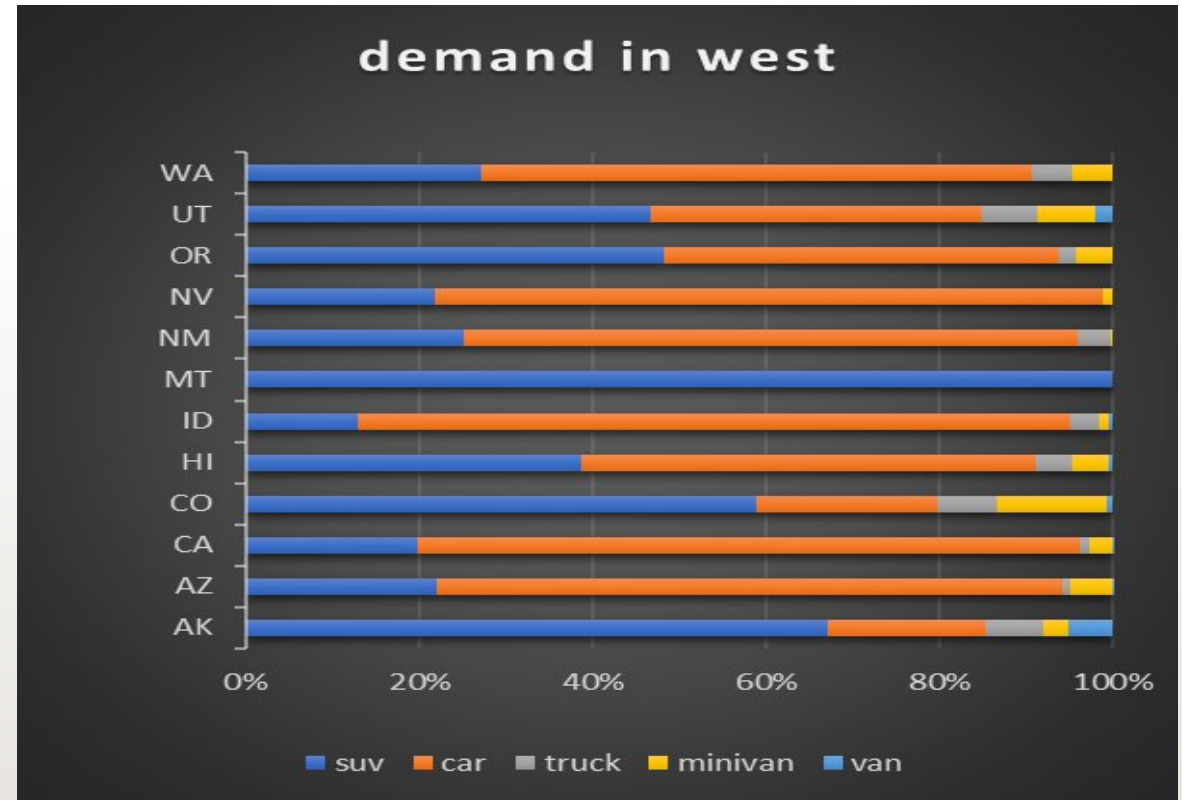


DEMAND OF DIFFERENT CAR TYPES IN DIFFERENT REGIONS

Montana, Colorado , Alaska has a significant demand for suv.

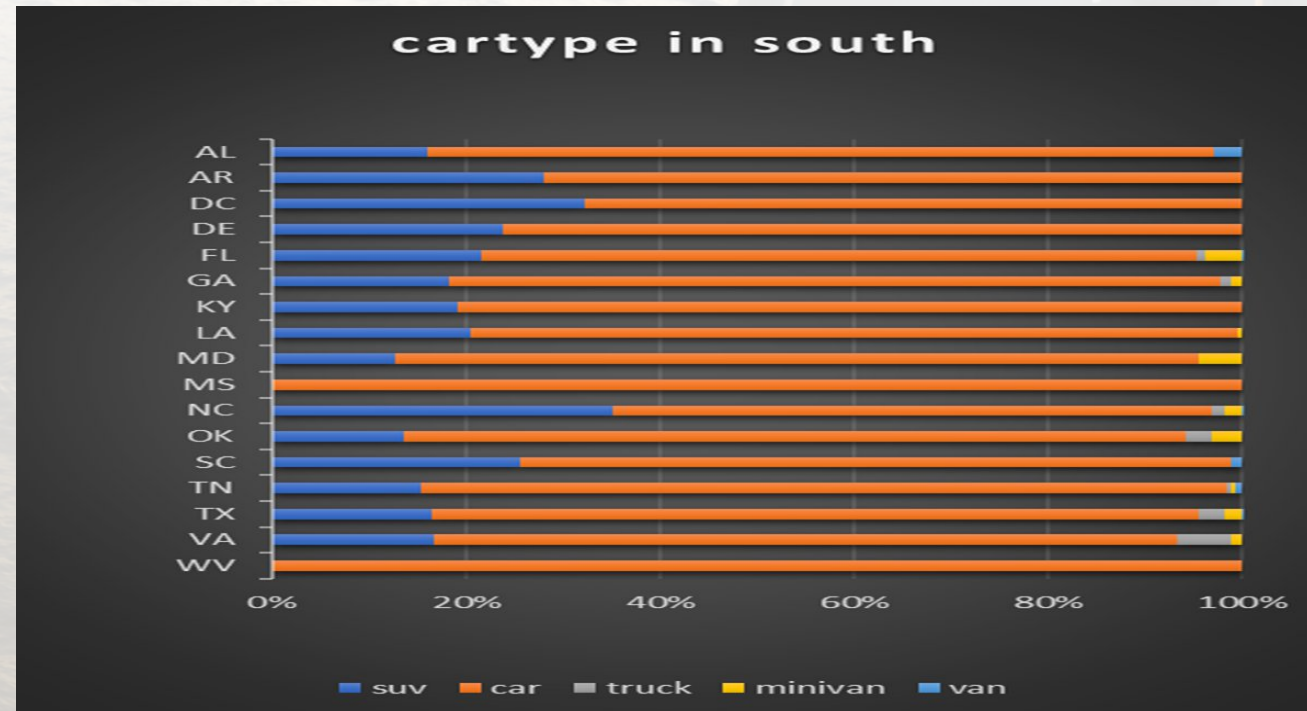
While Colorado , Utah and Oregon has demand for minivan.

And Alaska has potential for trucks.



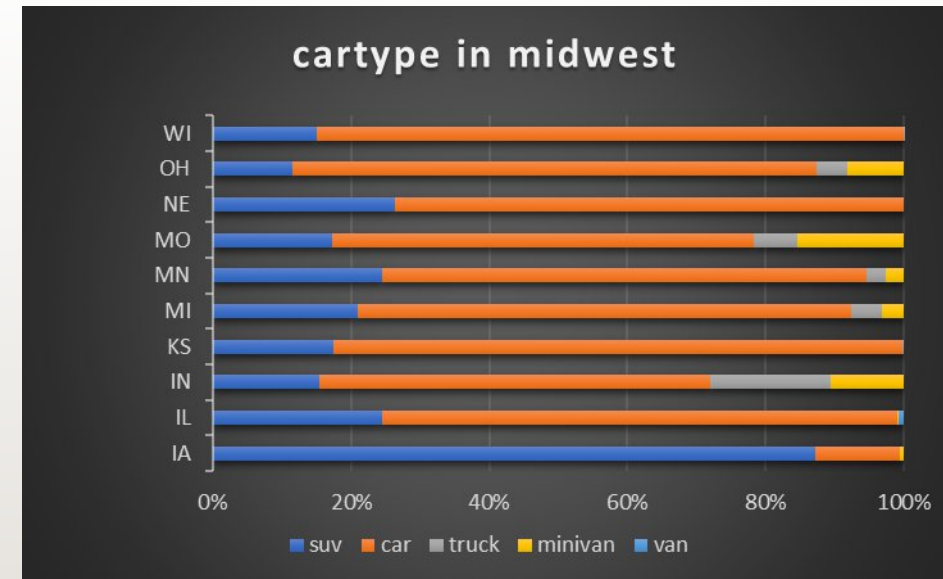
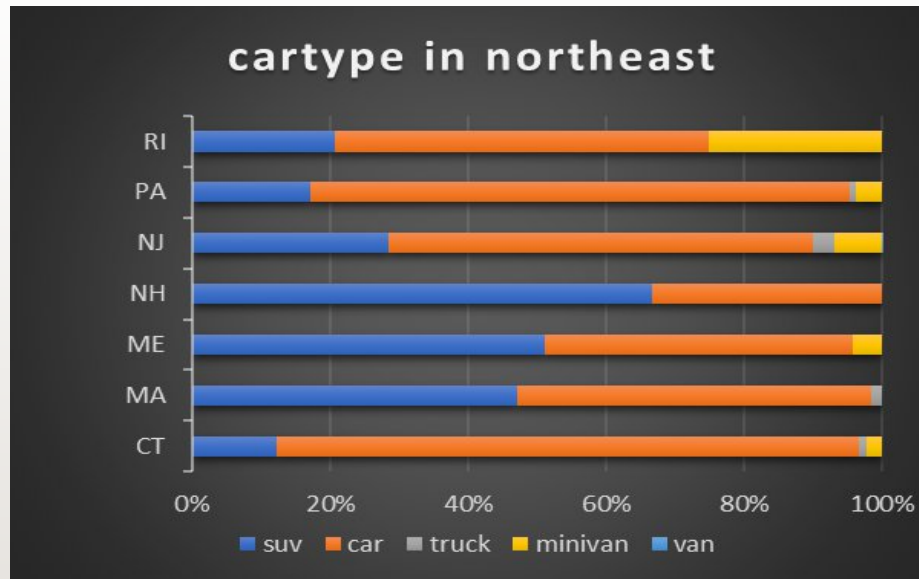
REGION SOUTH

Despite having second largest market share, region south does not have potential for any other type of cars except for cars and suvs.



Top Ranked Data Science & Analytics Education Provider since 2007

NORTHEAST AND MIDWEST

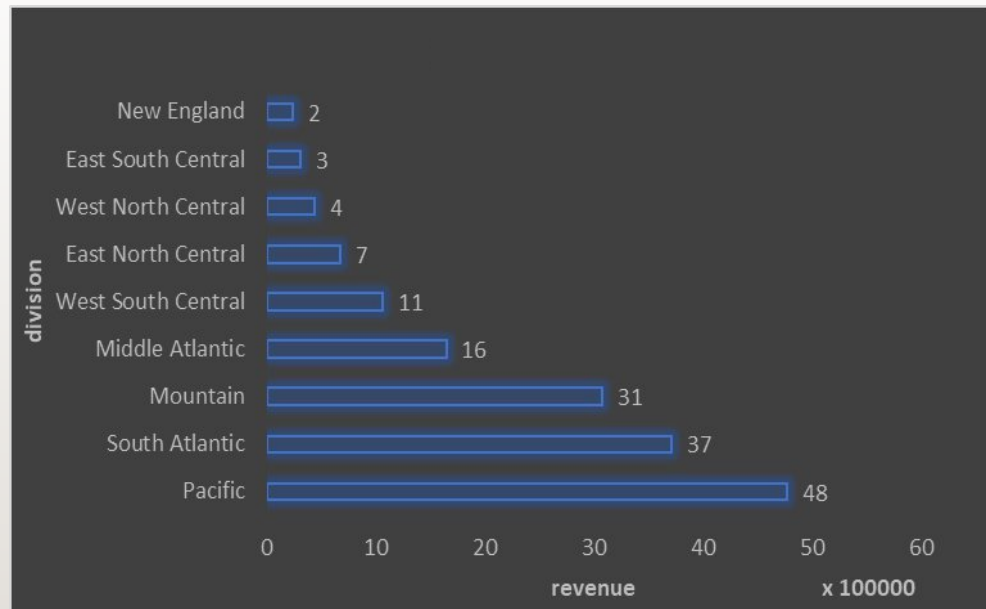


Top Ranked Data Science & Analytics Education Provider since 2007

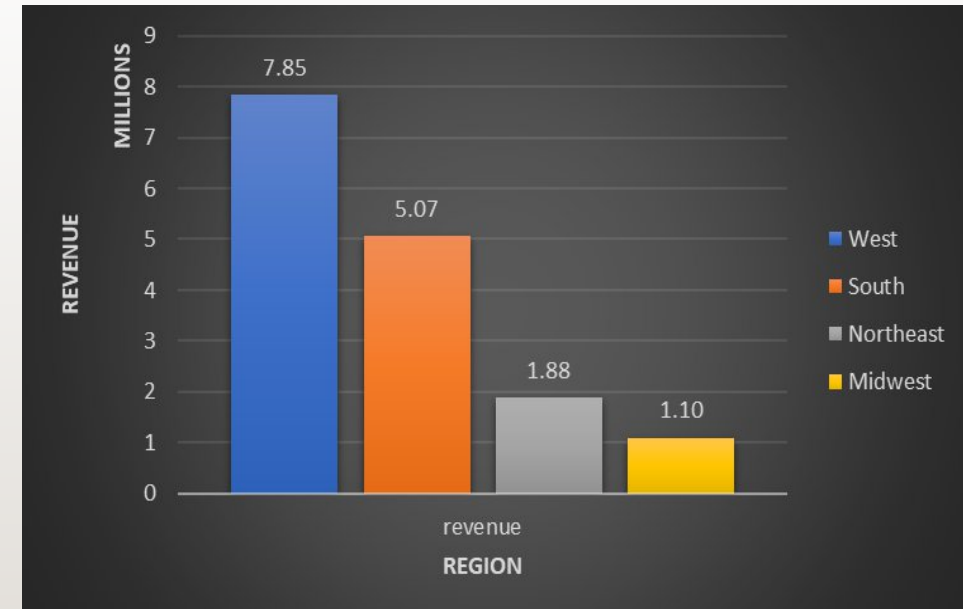
DIVISION AND REGION WISE REVENUE

COLLECTION

DIVISION WISE BUSINESS



REGION WISE BUSINESS

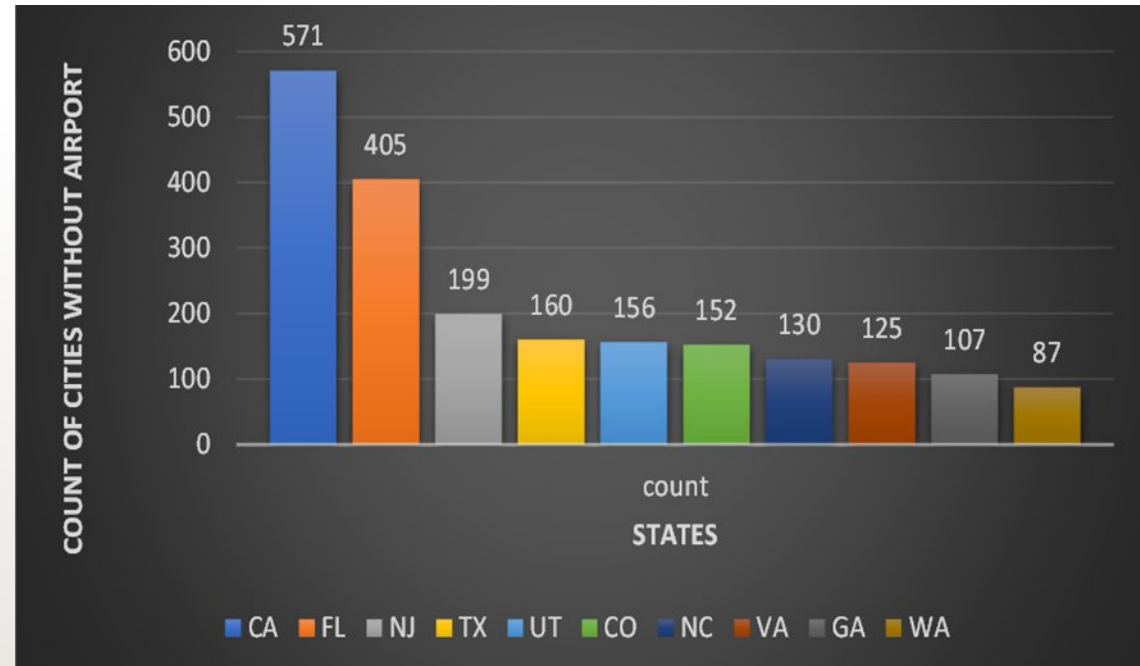


Top Ranked Data Science & Analytics Education Provider since 2007

OPPORTUNITIES FOR MARKET GROWTH

Top 10 cities where rental cars can be increased because there is no airport.

CA has the most cities without any airports followed by Florida and New Jersey.



KEY FINDINGS

- Toyota is the preferred choice of car. Tesla is close second followed by Chevrolet, BMW and Ford.
- Tesla earns the most revenue.
- 929 owner has more than one car invested while 15 people has more than 20 cars invested.
- California tops the list of most rentals followed by Florida and Texas.
- Vegas ,Jersey City, Los Angeles earns the most revenue.
- California, Florida, New Jersey has the most growth potential.
- West region has 1.5 time the market share of South region.
- sports cars are a big hit.
- Feedback is positive and ratings are genuine.



THANK YOU

ivy Professional School - Analytic Education Provider since 2007